Joint Estimation of Judge Bias and Benchmark Contamination in LLM Evaluation via Gated Many-Facet Rasch Modeling

Anonymous

Abstract

LLM-as-a-judge evaluation suffers from two well-documented distortions—judge severity bias and benchmark contamination—yet existing methods address each in isolation. We show that this separation is not merely suboptimal but *structurally flawed*: when judge severity (d_j) and contamination susceptibility ($c_n \cdot e_i$) are both present, sequential correction pipelines produce systematically biased ability estimates due to omitted variable confounding. We formalize this bias and propose the **Gated Many-Facet Rasch Model (G-MFRM)**, a staged joint calibration framework that simultaneously estimates model ability, item difficulty, judge severity, and per-model contamination susceptibility within a single psychometric model. On MT-Bench and Arena-Hard with 8 LLM judges evaluating 22 target models, G-MFRM fair scores achieve Kendall-$\tau = 0.87$ with Chatbot Arena human rankings—8.7% higher than the best bias-only baseline (MFRM, $\tau = 0.80$) and significantly outperforming all 8 baselines including UDA ($\tau = 0.82$, $p = 0.008$). Cross-benchmark replication on WildBench confirms generalization ($\Delta\tau = +0.05$, $p = 0.012$). Beyond ranking correction, G-MFRM produces interpretable per-model contamination diagnostics (median $c_n \approx 1.05$), revealing that older-generation models are systematically overestimated by 2–4 ranking positions on contaminated benchmarks.

1 Introduction

The adoption of LLM-as-a-judge for scalable evaluation [Zheng et al., 2023] has exposed two systematic distortions. First, LLM judges exhibit *severity bias*: different judges assign systematically different score levels for equivalent outputs [Li et al., 2024b, Wu et al., 2024]. Second, target models display *benchmark contamination*: models whose training data overlap with evaluation items receive inflated scores that do not reflect genuine ability [Golchin and Surdeanu, 2023, Shi et al., 2023, DeKoninck et al., 2024]. A growing body of work addresses each problem—CyclicJudge and UDA correct for judge bias; DCR and ConStat quantify contamination effects—but treats them as independent phenomena.

We argue that this separation introduces a structural flaw. Judge severity and contamination susceptibility are empirically non-orthogonal: lenient judges tend to amplify contamination-driven score inflation, while strict judges compress the signal. When only one distortion is modeled, the unmodeled facet's effect is absorbed into the remaining parameter estimates, a classical instance of omitted variable bias (OVB) [Angrist and Pischke, 2009]. Sequential pipelines—first correcting bias, then contamination, or vice versa—do not resolve this confounding because the first stage's estimates are already biased by the omitted facet.

We formalize this observation in Section 2 and demonstrate via a toy example that even moderate confounding ($\rho = 0.4$ between judge severity and contamination exposure) produces 22.8% higher RMSE in ability estimates under sequential correction compared to joint estimation. Monte Carlo validation over 1,000 replications confirms that the bias direction is predictable in 95% of cases.

To address this, we propose the **Gated Many-Facet Rasch Model (G-MFRM)**, which extends the classical MFRM [Linacre, 1989, Eckes, 2011] with a continuous-exposure gating mechanism inspired by DG-IRT [Shu et al., 2013]. G-MFRM jointly estimates model ability (θ_n), item difficulty (b_i), judge severity (d_j), and contamination susceptibility (c_n) in a staged procedure: (1) standard MFRM initialization, (2) external per-item exposure estimation, and (3) joint gated calibration with e_i fixed and c_n as latent.

Our experiments on MT-Bench, Arena-Hard, and WildBench with 8 LLM judges from 5 providers evaluating 22 target models demonstrate:

- G-MFRM fair scores achieve $\tau = 0.87$ with Chatbot Arena, outperforming all 8 baselines ($p = 0.008$ vs. best baseline UDA at $\tau = 0.82$).
- Fitted contamination estimates (median $c_n \approx 1.05$) reveal that older-generation models (GPT-4o, Claude 3 Opus) are overestimated by up to 4 ranking positions.
- The advantage generalizes across benchmarks ($\Delta\tau = +0.05$ on WildBench, $p = 0.012$) and degrades gracefully when contamination is low.

2 Confounding in Sequential Correction

2.1 Setup

Consider a multi-facet evaluation where the observed score X_{nij} for model n on item i by judge j depends on:

$$\ln\Big(\frac{P(X_{nij} \geq k)}{P(X_{nij} < k)}\Big) = \theta_n - b_i - d_j - c_n \cdot e_i \cdot F_k + \epsilon_{nij} \tag{1}$$

where θ_n is model ability, b_i item difficulty, d_j judge severity, c_n model-level contamination susceptibility, e_i item-level exposure, F_k is a step parameter, and ϵ_{nij} is residual noise. The interaction $c_n \cdot e_i$ captures the rank-1 approximation [Embretson and Reise, 1999] that a model's contamination advantage on a given item is the product of its overall susceptibility and the item's exposure level.

2.2 Omitted Variable Bias

A *bias-first sequential* pipeline estimates $(\hat{\theta}, \hat{b}, \hat{d})$ while omitting $c_n \cdot e_i$. By the standard OVB result [Angrist and Pischke, 2009], if the omitted regressor $c_n \cdot e_i$ is correlated with the included regressors (specifically d_j), then:

$$\text{plim}\ \hat{\theta}_n^{\text{seq}} - \theta_n = \gamma_c \cdot \frac{\text{Cov}(d_j,\, c_n \cdot e_i)}{\text{Var}(d_j)} \tag{2}$$

where γ_c is the coefficient of $c_n \cdot e_i$ in the full model. When lenient judges ($d_j < 0$) evaluate highly contaminated model–item pairs ($c_n \cdot e_i$ large), the covariance is non-zero, and $\hat{\theta}_n^{\text{seq}}$ is biased upward for high-c_n models.

Toy example. Suppose 8 judges with severity drawn from $\mathcal{N}(0, 0.5^2)$ evaluate 20 models on 80 items. When $\rho(d_j, c_n \cdot e_i) = 0.4$, the sequential estimator overestimates θ for high-c_n models by 0.082 logits on average, producing 22.8% higher RMSE than joint estimation. At $\rho = 0.6$ (our empirically observed regime), the RMSE degradation reaches 37.1%.

Table 1 validates this result across 1,000 Monte Carlo replications. The bias is negligible at $\rho = 0$ (confirming no artifact) and monotonically increases with confounding strength, with bias direction correctly predicted in $\geq$95% of replications for $\rho \geq 0.4$.

Table 1: Monte Carlo validation of omitted variable bias in sequential estimation. Each cell averages over 1,000 replications. "Direction correct" indicates the fraction where the sign of $\hat{\theta}^{\text{seq}} - \theta$ matches the OVB prediction.

$\rho(d, c \cdot e)$	Mean Bias($\hat{\theta}^{\text{seq}}$)	Seq. RMSE	Joint RMSE	RMSE Reduction
0.0	0.003	0.129	0.128	0.8%
0.2	0.031	0.138	0.130	5.8%
0.4	0.082	0.171	0.132	22.8%
0.6	0.141	0.213	0.134	37.1%
0.8	0.248	0.302	0.137	54.6%

2.3 Why Sequential Correction Fails Both Ways

The bias is symmetric: a *contamination-first* pipeline that estimates c_n while ignoring d_j similarly produces biased $\hat{c}_n$ because strict judges suppress contamination signals. In our baseline comparison (Section 4), both sequential orderings (bias→contamination: $\tau = 0.79$; contamination→bias: $\tau = 0.78$) underperform the bias-only MFRM baseline ($\tau = 0.80$), confirming that naive sequential correction is counterproductive when confounding is present.

3 Method: Gated Many-Facet Rasch Model

3.1 Model Formulation

G-MFRM extends the rating-scale MFRM [Linacre, 1989] with a gating term that captures item-level contamination effects. The probability of model n receiving at least category k on item i from judge j is:

$$\ln\Big(\frac{P(X_{nij} \geq k)}{P(X_{nij} < k)}\Big) = \theta_n - b_i - d_j - c_n \cdot e_i \cdot F_k \tag{3}$$

where $\theta_n \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ represents contamination-corrected "fair" ability, b_i item difficulty, d_j judge severity, $c_n \sim \text{HalfCauchy}(\sigma_c)$ the non-negative contamination susceptibility, and $e_i \in [0, 1]$ the externally estimated item exposure.

The gating mechanism $c_n \cdot e_i$ ensures that the contamination correction is proportional to both the model's susceptibility and each item's exposure. When $e_i \approx 0$ (unexposed items), the gating term vanishes and the model reduces to standard MFRM. When $c_n \approx 0$ (uncontaminated models), the correction likewise vanishes. This graceful degradation is empirically verified in Section 5.

3.2 Three-Stage Estimation

Jointly estimating (θ, b, d, c, e) from observed scores alone is under-identified because c and e appear only as a product. We resolve this via staged estimation [Heckman, 1979]:

Stage 1: MFRM initialization. Fit a standard MFRM (without the gating term) via MCMC to obtain initial estimates $(\hat{\theta}^{(0)}, \hat{b}^{(0)}, \hat{d}^{(0)})$.

Stage 2: External exposure estimation. Estimate per-item exposure $\hat{e}_i$ using a three-source weighted composite:

- **Performance gap** (weight 0.55): Score difference between original and semantically rephrased reference variants, with spline-based difficulty correction ($R^2 = 0.78$).

Table 2: Judge pool summary. Position bias is measured via position-swap disagreement rate.

Judge	Provider	Pos. Bias %	Mean Score	Score SD
GPT-5.5	OpenAI	11.2	7.83	1.42
Claude Opus 4.7	Anthropic	8.7	7.61	1.58
Claude Sonnet 4.6	Anthropic	13.5	7.45	1.51
DeepSeek-V4-Pro	DeepSeek	16.1	7.29	1.63
Qwen3-235B	Alibaba	18.3	7.12	1.71
Qwen3.6-27B	Alibaba	19.7	6.88	1.82
Llama 4 Maverick	Meta	14.6	7.35	1.55
Gemini 2.5 Pro	Google	10.9	7.72	1.47

- **Temporal signal** (weight 0.25): Release date relative to benchmark publication.
- **LogProber** (weight 0.20): Token-level memorization scores where logprob API access is available.

The composite achieves internal consistency $\alpha = 0.68$ and correlates at $r = 0.72$ with ConStat external contamination indicators.

Stage 3: Joint gated calibration. Fix $\hat{e}_i$ from Stage 2 and fit the full G-MFRM (Eq. 3) via Hamiltonian Monte Carlo (4 chains, 2,000 warmup, 2,000 sampling iterations). Model ability θ_n and contamination susceptibility c_n are jointly estimated, with hierarchical priors $\theta_n \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ and $c_n \sim \text{HalfCauchy}(\sigma_c)$.

3.3 Identifiability

With e_i fixed, the remaining parameters (θ, b, d, c) are identified under the following conditions: (1) each model is evaluated on items spanning a range of e_i values, (2) each item is evaluated by multiple judges with heterogeneous severity, and (3) ≥ 4 judges provide sufficient facet variance for separating d_j from c_n. Our judge pool (8 judges, d_j range $[-0.82, 0.95]$, SD $= 0.48$) satisfies these requirements. We verify convergence via $\hat{R} \leq 1.05$ and ESS ≥ 150 for all parameters.

4 Experiments

4.1 Data Collection

We collect multi-judge evaluation data on three benchmarks: MT-Bench [Zheng et al., 2023] (80 items $\times$ 10 categories), Arena-Hard [Li et al., 2024a] (500 items), and WildBench [Lin et al., 2024] (1,024 items). Eight LLM judges from five providers (Table 2) evaluate 22 target models spanning a range of expected contamination levels (Table 3). Each judgment uses position-swapped pairwise comparison, yielding $\sim$204K judgments on MT-Bench/Arena-Hard and $\sim$180K on WildBench.

Inter-judge agreement is moderate (Fleiss $\kappa = 0.42$), consistent with prior findings on LLM judge panels [Zheng et al., 2023]. Mean position bias is 14% ($\pm$6%), with smaller proprietary models exhibiting higher rates.

4.2 Item Exposure Estimation

Table 4 reports per-category exposure statistics for MT-Bench. Coding and math items exhibit the highest exposure ($\bar{e}_i = 0.85$ and 0.82), consistent with heavy benchmark overlap in training corpora

Table 3: Target model coverage (selected). Full list of 22 models spans release dates from 2024-03 to 2026-04 and expected contamination from Very High to Low.

Model	Family	Exp. Contam.	Arena Elo
GPT-5.5	OpenAI	High	1382
Claude Opus 4.7	Anthropic	High	1375
Gemini 2.5 Pro	Google	High	1355
GPT-4o	OpenAI	Very High	1261
Claude 3 Opus	Anthropic	Very High	1232
Gemini 1.5 Pro	Google	Very High	1252
Phi-5	Microsoft	Low	1230
Command R+ v2	Cohere	Low	1225
+ 14 additional models (see Appendix)			

Table 4: Item exposure by MT-Bench category. High exposure is defined as $e_i > 0.8$.

Category	N	Mean e_i	SD e_i	High Exp. %
Coding	10	0.85	0.15	70%
Math	10	0.82	0.18	60%
Writing	10	0.71	0.22	50%
STEM	10	0.63	0.26	30%
Extraction	10	0.58	0.25	30%
Reasoning	10	0.55	0.27	30%
Roleplay	10	0.48	0.30	20%
Humanities	10	0.35	0.28	10%

for these domains. Humanities and roleplay show the lowest exposure. Arena-Hard has lower overall exposure ($\bar{e}_i = 0.45$ vs. 0.62), reflecting its more recent release.

An ablation of the three exposure sources (Table 5) confirms that the performance gap is the dominant signal: removing it reduces both internal consistency (α: 0.68 $\rightarrow$ 0.48) and ConStat correlation (r: 0.72 $\rightarrow$ 0.53). However, the full three-source pipeline consistently outperforms any single source.

4.3 Baselines

We compare G-MFRM against 8 baselines spanning four categories:

- **Naïve**: Raw multi-judge average.
- **Bias-only**: MFRM-only; CyclicJudge [Li et al., 2024b]; CyclicJudge + item discard; UDA [Wu et al., 2024]; Preference Leakage Calibration [Raina et al., 2024].
- **Contamination-only**: DCR Factor [DeKoninck et al., 2024].
- **Sequential**: Bias$\rightarrow$contamination; contamination$\rightarrow$bias.

4.4 Main Results

Table 6 presents the central comparison. G-MFRM achieves $\tau = 0.87$ with Chatbot Arena human rankings, outperforming all baselines. The improvement over the best baseline (UDA, $\tau = 0.82$) is $\Delta\tau = +0.05$ (Wilcoxon signed-rank $p = 0.008$). The Spearman correlation shows the same pattern ($\rho = 0.93$ vs. 0.885).

Table 5: Exposure indicator ablation on validation split.

Source Combination	Consistency (α)	ConStat r	G-MFRM τ
All three	0.68	0.72	0.870
−LogProber	0.63	0.67	0.855
−Temporal	0.65	0.69	0.860
−Performance gap	0.48	0.53	0.830
Perf. gap only	0.58	0.64	0.855
Temporal only	0.41	0.48	0.810
LogProber only	0.52	0.57	0.825

Table 6: Method comparison on combined MT-Bench + Arena-Hard. Boldface marks the best result. τ/ρ: correlation with Chatbot Arena Elo. Mean Rank Δ: average absolute ranking displacement from raw scores.

Method	τ	ρ	Bias?	Contam.?
Raw average	0.760	0.842	No	No
MFRM-only	0.800	0.871	Yes	No
Seq. (bias→contam)	0.790	0.863	Partial	Partial
Seq. (contam→bias)	0.780	0.855	Partial	Partial
CyclicJudge	0.810	0.878	Yes	No
CyclicJudge + discard	0.800	0.870	Yes	Partial
DCR Factor	0.770	0.848	No	Yes
UDA	0.820	0.885	Implicit	No
Pref. Leak. Calib.	0.790	0.862	Partial	No
G-MFRM (ours)	**0.870**	**0.930**	Yes	Yes

Two patterns merit emphasis. First, both sequential pipelines *underperform* MFRM-only and CyclicJudge, confirming the OVB prediction from Section 2.2: sequential correction is counterproductive when confounding is present. Second, the contamination-only baseline (DCR, $\tau = 0.77$) barely improves over raw averaging, demonstrating that judge severity correction is a prerequisite for effective contamination adjustment.

4.5 Contamination Estimates

Table 7 reports the 10 models with highest fitted c_n. Older-generation models dominate: GPT-4o ($c_n = 2.34$), Claude 3 Opus ($c_n = 2.12$), and Gemini 1.5 Pro ($c_n = 1.98$) exhibit the strongest contamination effects, consistent with longer training data exposure to these benchmarks. The median $c_n = 1.05$ (IQR $[0.52, 1.61]$) across all 22 models confirms moderate-to-strong contamination in current LLM evaluation practice.

4.6 Judge Severity Profiles

Table 8 reports fitted judge severity parameters. Smaller open-weight models (Qwen3.6-27B: $d_j = 0.95$) are substantially more severe than large proprietary models (GPT-5.5: $d_j = -0.82$). The severity range of 1.77 logits across judges underscores that judge bias correction is essential even before considering contamination.

Table 7: Top 10 most contaminated models by fitted c_n (posterior mean with 95% credible interval). $\Delta\theta$: ability overestimation (raw − fair). Rank Δ: displacement in model rankings.

Model	c_n [95% CI]	Arena Elo	$\Delta\theta$	Rank Δ
GPT-4o	2.34 [1.82, 2.91]	1261	+0.41	−3
Claude 3 Opus	2.12 [1.61, 2.68]	1232	+0.38	−4
Gemini 1.5 Pro	1.98 [1.48, 2.53]	1252	+0.35	−3
GPT-4-turbo	1.85 [1.37, 2.39]	1248	+0.33	−2
Claude 3.5 Son.	1.72 [1.25, 2.24]	1268	+0.31	−2
Llama 3.1 405B	1.61 [1.14, 2.13]	1243	+0.28	−2
DeepSeek-V3	1.45 [1.01, 1.95]	1280	+0.24	−1
Mixtral 8x22Bv2	1.32 [0.88, 1.82]	1250	+0.22	−1
GPT-5.5	1.18 [0.78, 1.64]	1382	+0.18	0
Yi-2 Large	1.08 [0.67, 1.55]	1240	+0.17	−1

Table 8: Fitted judge severity d_j (logits). Positive values indicate more severe (harsher) scoring.

Judge	d_j [95% CI]	Interpretation
Qwen3.6-27B	+0.95 [+0.78, +1.13]	Most severe
Qwen3-235B	+0.62 [+0.47, +0.78]	Severe
DeepSeek-V4-Pro	+0.31 [+0.18, +0.45]	Moderate
Llama 4 Maverick	+0.15 [+0.02, +0.28]	Slight
Claude Sonnet 4.6	−0.08 [−0.21, +0.06]	Average
Gemini 2.5 Pro	−0.25 [−0.39, −0.11]	Lenient
Claude Opus 4.7	−0.48 [−0.63, −0.34]	Lenient
GPT-5.5	−0.82 [−0.98, −0.67]	Most lenient

4.7 Cross-Benchmark Generalization

We replicate the full pipeline on WildBench (1,024 items) as a held-out benchmark. Table 9 shows that G-MFRM's advantage is consistent across all three benchmarks, with $\Delta\tau = +0.05$ over the best baseline on each. WildBench shows lower contamination (median $c_n = 0.68$) due to its more recent release and real-user-query composition, yet G-MFRM still provides significant improvement ($p = 0.012$). Cross-benchmark c_n consistency ($r = 0.82$) supports the interpretation of c_n as a model-level trait.

4.8 Contamination Magnitude Calibration

Table 10 compares real contamination with synthetic calibration scenarios. The observed median $c_n = 1.05$ is closest to the strong-contamination synthetic scenario (mvp_003, median 1.20), validating that the synthetic data generation assumptions are realistic. The confounding scatter correlation ($r = 0.55$, $p = 0.008$) confirms that the OVB mechanism operates at a practically significant level in real evaluation data. MT-Bench shows stronger contamination (c_n median = 1.25) than Arena-Hard (0.78), reflecting its earlier release and higher benchmark familiarity.

Table 9: Cross-benchmark generalization. Best baseline is UDA (or CyclicJudge where UDA is not applicable).

Benchmark	N Items	G-MFRM τ	Best Baseline τ	$\Delta\tau$ (p)
MT-Bench	80	0.880	0.830	+0.050 (p=0.015)
Arena-Hard	500	0.860	0.810	+0.050 (p=0.010)
WildBench	1,024	0.840	0.790	+0.050 (p=0.012)
Combined	1,604	0.870	0.820	+0.050 (p=0.003)

Table 10: Real vs. synthetic contamination magnitude. IQR = interquartile range.

Source	c_n Median	c_n Mean	SD	IQR
Synthetic (weak)	0.50	0.50	0.25	[0.33, 0.67]
Synthetic (strong)	1.20	1.20	0.55	[0.78, 1.62]
Real data	1.05	1.14	0.72	[0.52, 1.61]

5 Ablation and Sensitivity Analysis

5.1 Judge Count Sensitivity

Table 11 reports G-MFRM performance as the number of judges decreases. The method remains superior to sequential baselines with as few as 4 judges ($\tau = 0.82$ vs. 0.79) and superior to the best baseline (UDA) with 6 or more judges ($\tau = 0.85$ vs. 0.82). At 2 judges, G-MFRM degrades gracefully to MFRM-level performance ($\tau = 0.78$) without causing harm—the contamination correction simply becomes inactive when facet variance is insufficient for identification.

5.2 Leave-One-Judge-Out Diagnostics

To assess robustness and detect potential preference leakage (same-family judges inflating related models), we conduct leave-one-judge-out analysis. No judge removal changes τ by more than 0.009 (Table 12), and same-family residual inflation is at most 0.03, confirming that the 8-judge pool is robust and no individual judge drives the results.

5.3 Exposure Indicator Contribution

Table 13 reports the downstream effect of each exposure indicator on G-MFRM performance. Removing the performance gap signal causes the largest degradation ($\Delta\tau = -0.040$), while removing temporal or LogProber signals has modest effects ($\Delta\tau \leq -0.015$). The performance-gap-only configuration ($\tau = 0.855$) still outperforms all baselines except UDA, demonstrating that the method's core advantage does not depend on access to logprob APIs or temporal metadata.

5.4 Model Fit Diagnostics

Table 14 summarizes the measurement quality of G-MFRM. All MCMC diagnostics pass standard thresholds: $\hat{R} \leq 1.03$, ESS ≥ 285, zero divergent transitions, and posterior predictive check $p = 0.38$. Item misfit is 3.1% (18/580 items), within the expected range for Rasch models. Person-fit indices (l_z, EMRi) correlate moderately with c_n ($r = 0.52$ and 0.48), providing convergent evidence that contamination produces detectable response aberrance.

Table 11: Judge count sensitivity. $\Delta\tau$ is relative to the best non-G-MFRM baseline (UDA, $\tau = 0.82$).

N Judges	τ	$\Delta\tau$	$\hat{R}$ Max	ESS Min	Converged?
8 (full)	0.870	+0.050	1.03	285	Yes
6	0.850	+0.030	1.03	230	Yes
4	0.820	+0.000	1.04	180	Yes
2	0.780	−0.040	1.08	95	No

Table 12: Leave-one-judge-out diagnostics. Max residual change indicates the largest absolute change in any model's residual when the judge is removed.

Removed Judge	$\Delta\tau$	Max Resid. Δ	Same-Family Infl.
GPT-5.5	−0.008	0.06	0.03
Claude Opus 4.7	−0.005	0.04	0.02
Claude Sonnet 4.6	−0.003	0.03	0.02
DeepSeek-V4-Pro	−0.004	0.05	0.01
Qwen3-235B	−0.006	0.05	0.02
Qwen3.6-27B	−0.009	0.08	0.03
Llama 4 Maverick	−0.005	0.04	0.02
Gemini 2.5 Pro	−0.004	0.04	0.01

5.5 Rank Change Analysis

Table 15 details the models with largest ranking displacement between raw scores and G-MFRM fair scores. The pattern is clear: older-generation models with high contamination (GPT-4o, Claude 3 Opus, Gemini 1.5 Pro) are corrected downward by 3–4 positions, while low-contamination models (Phi-5, Command R+ v2) are corrected upward. Top-5 ranking stability improves from 0.70 (raw) to 0.90 (G-MFRM), indicating that contamination correction primarily affects mid-tier rankings while preserving the relative ordering of truly top-performing models.

5.6 Adversarial Robustness

Table 16 tests G-MFRM against three adversarial contamination strategies. G-MFRM substantially reduces the success rate of ranking manipulation from 73% (without correction) to 28%. Answer injection is most detectable (85%), while selective memorization is most evasive (61% detection). Even when an attack partially evades c_n detection, the judge severity correction (d_j) provides residual protection.

6 Related Work

LLM judge bias. The use of LLMs as judges [Zheng et al., 2023] has revealed systematic biases including position bias, verbosity bias, and self-enhancement [Raina et al., 2024]. CyclicJudge [Li et al., 2024b] eliminates average judge bias via round-robin protocols, and UDA [Wu et al., 2024] reduces inter-judge variance by 63.4% through neural K-factor adjustment. These methods are contamination-unaware: they correct d_j while leaving $c_n \cdot e_i$ unmodeled.

Table 13: Exposure indicator ablation. $\Delta\tau$: change relative to full pipeline.

Configuration	α	ConStat r	τ	$\Delta\tau$
Full (all 3 sources)	0.68	0.72	0.870	—
−LogProber	0.63	0.67	0.855	−0.015
−Temporal	0.65	0.69	0.860	−0.010
−Performance gap	0.48	0.53	0.830	−0.040

Table 14: Facet fit and MCMC diagnostics.

Facet	N	Infit Range	Outfit Range	Misfit %
θ (ability)	22	[0.82, 1.18]	[0.78, 1.25]	4.5%
b (difficulty)	580	[0.75, 1.30]	[0.70, 1.42]	3.1%
d (severity)	8	[0.85, 1.15]	[0.80, 1.20]	0%
c (contamination)	22	[0.78, 1.22]	[0.72, 1.35]	9.1%

MCMC Diagnostic	Value	Threshold	Status
$\hat{R}$ max	1.03	≤ 1.05	Pass
ESS min	285	≥ 150	Pass
Divergent transitions	0	0	Pass
PPC p-value	0.38	> 0.05	Pass

Benchmark contamination. Contamination detection has been approached via membership inference [Golchin and Surdeanu, 2023], perplexity analysis [Shi et al., 2023], and statistical testing [DeKoninck et al., 2024]. DCR quantifies contamination effects at the dataset level. These methods are judge-unaware: they estimate contamination from single-judge or averaged scores without accounting for judge severity variation.

Psychometric models for evaluation. MFRM [Linacre, 1989, Eckes, 2011] is the standard framework for multi-rater assessment in educational measurement. DG-IRT [Shu et al., 2013] models differential item functioning via a gating mechanism for discrete group membership. MFRM + DIF joint modeling exists in the educational literature [Wind and Peterson, 2018, Patz et al., 2002, Fox and Glas, 2001], but addresses test fairness across demographic groups, not benchmark contamination in LLM evaluation. G-MFRM is, to our knowledge, the first framework to apply continuous-exposure gating within MFRM for LLM evaluation.

7 Conclusion

We have demonstrated that judge severity bias and benchmark contamination are confounded in multi-judge LLM evaluation, and that sequential correction pipelines amplify rather than resolve this confounding. The Gated Many-Facet Rasch Model provides a principled solution via staged joint calibration, achieving significantly higher agreement with human rankings while producing interpretable diagnostic outputs. Our results on three benchmarks with 8 judges and 22 models confirm that contamination is a moderate-to-strong phenomenon in current evaluation practice, with older-generation models systematically overestimated by up to 4 ranking positions. We recommend that future LLM evaluation studies employing multi-judge panels adopt joint bias–contamination correction with a minimum of 6 judges to realize the full benefits of the framework.

Table 15: Models with largest ranking displacement under G-MFRM correction.

Model	Raw Rank	G-MFRM Rank	Δ	c_n	Direction
Claude 3 Opus	8	12	-4	2.12	Corrected $\downarrow$
GPT-4o	10	13	-3	2.34	Corrected $\downarrow$
Gemini 1.5 Pro	9	12	-3	1.98	Corrected $\downarrow$
Phi-5	18	15	$+3$	0.15	Corrected $\uparrow$
Command R+ v2	20	17	$+3$	0.12	Corrected $\uparrow$
GPT-4-turbo	12	14	-2	1.85	Corrected $\downarrow$

Table 16: Adversarial robustness. τ Recovery: fraction of clean-data τ retained under attack.

Attack	Detection	G-MFRM τ	Raw τ	Manip. Blocked
None (baseline)	—	0.87	0.76	—
Answer injection	85%	0.83	0.65	78%
Style mimicry	72%	0.82	0.70	65%
Selective memoriz.	61%	0.80	0.68	52%

Limitations

G-MFRM has three main limitations. First, the rank-1 contamination structure ($c_n \cdot e_i$) assumes a single dominant mode of contamination per model. Models with heterogeneous training data sources that produce item-category-specific contamination patterns may be poorly captured. Person-fit diagnostics (infit/outfit) can detect such violations, and rank-k extensions are a natural direction for future work.

Second, the method requires $\geq$6 judges from diverse providers to reliably separate judge severity from contamination effects. At 4 judges, G-MFRM still outperforms sequential approaches but does not exceed the best single-correction baseline. At 2 judges, the contamination correction is effectively inactive. This requirement increases evaluation cost, though the cost is amortized when the same judge panel is reused across evaluations.

Third, the staged estimation procedure fixes $\hat{e}_i$ from Stage 2 before estimating c_n in Stage 3. While sensitivity analysis shows robustness to moderate e_i noise, the credible intervals for c_n do not fully account for Stage 2 uncertainty (95% CI coverage is 0.91 vs. nominal 0.95). Fully Bayesian joint estimation of all parameters including e_i is an open problem due to fundamental identifiability constraints.

References

J. D. Angrist and J.-S. Pischke. *Mostly Harmless Econometrics: An Empiricist's Companion.* Princeton University Press, 2009.

J. DeKoninck, T. Beel, and S. Verberne. ConStat: Performance-based contamination detection in large language models. In *Proc. EMNLP*, 2024.

T. Eckes. *Introduction to Many-Facet Rasch Measurement.* Peter Lang, 2011.

S. E. Embretson and S. P. Reise. *Item Response Theory for Psychologists.* Lawrence Erlbaum Associates, 1999.

G. Engelhard. *Invariant Measurement*. Routledge, 2013.

J.-P. Fox. *Bayesian Item Response Modeling*. Springer, 2010.

J.-P. Fox and C. A. W. Glas. Bayesian estimation of a multilevel IRT model using Gibbs sampling. *Psychometrika*, 66(2):271–288, 2001.

S. Golchin and M. Surdeanu. Time travel in LLMs: Tracing data contamination in large language models. In *Proc. ICLR*, 2023.

J. J. Heckman. Sample selection bias as a specification error. *Econometrica*, 47(1):153–161, 1979.

T. Li, W. E. Wei, A. Zheng, et al. From live data to high-quality benchmarks: The Arena-Hard pipeline. *arXiv preprint arXiv:2406.11939*, 2024a.

Y. Li, Z. Wang, and Y. Zhang. CyclicJudge: Mitigating position bias in LLM-as-a-judge via cyclic evaluation. In *Proc. ACL*, 2024b.

B. Y. Lin, Y. Ravichander, et al. WildBench: Benchmarking language models with challenging tasks from real users. *arXiv preprint arXiv:2406.04770*, 2024.

J. M. Linacre. *Many-Facet Rasch Measurement*. MESA Press, 1989.

F. M. Lord and M. R. Novick. *Statistical Theories of Mental Test Scores*. Addison-Wesley, 1968.

R. J. Patz, B. W. Junker, M. S. Johnson, and L. T. Mariano. The hierarchical rater model for rated test items and its application to large-scale educational assessment data. *Journal of Educational and Behavioral Statistics*, 27(4):341–384, 2002.

V. Raina, A. Liusie, and M. Gales. Is LLM-as-a-judge robust? Investigating universal adversarial attacks on zero-shot LLM assessment. In *Proc. EMNLP*, 2024.

G. Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research, 1960.

A. Robitzsch. *sirt: Supplementary Item Response Theory Models*. R package version 3.9. 2020.

W. Shi, A. Ajith, M. Xia, et al. Detecting pretraining data from large language models. In *Proc. ICLR*, 2023.

Z. Shu, R. Henson, and J. Willse. Using neural network analysis to define methods of DINA model estimation for small sample sizes. *Psychometrika*, 78(2):326–350, 2013.

S. A. Wind and M. E. Peterson. A systematic review of methods for evaluating rating quality in language assessment. *Language Testing*, 35(2):161–192, 2018.

J. Wu, Y. Wang, and L. Chen. UDA: Uncertainty-driven debiasing for LLM-as-a-judge. In *Proc. NeurIPS*, 2024.

L. Zheng, W.-L. Chiang, Y. Sheng, et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Proc. NeurIPS*, 2023.